

B.M.S. COLLEGE OF ENGINEERING

(Autonomous Institute, Affiliated to VTU)

Bull Temple Road, Basavanagudi, Bengaluru - 560019



A Capstone Project Report on

**“DRYRUNAI: A GAN-CENTRIC GENERATIVE AI
FRAMEWORK FOR PRE-LAUNCH PRODUCT SIMULATION AND
PRODUCT-MARKET FIT OPTIMIZATION”**

Submitted in partial fulfilment of the requirements for the award of degree

BACHELOR OF ENGINEERING

IN

INFORMATION SCIENCE AND ENGINEERING

By

Samanyu Gupta — 1BM23IS216

Sarvagnya Akamanchi — 1BM23IS220

Sejil Nooh Ansari — 1BM23IS221

Shlok Naik Charan — 1BM23IS229

Under the guidance of

Dr. Shobha T
Associate Professor

Department of Information Science and Engineering
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Bengaluru – 560019

Department of Information Science and Engineering

CERTIFICATE

This is to certify that the project entitled “**DryRunAI: A GAN-Centric Generative AI Framework for Pre-Launch Product Simulation and Product–Market Fit Optimization**” is a bona-fide work carried out by **Samanyu Gupta (1BM23IS216)**, **Sarvagnya Akamanchi (1BM23IS220)**, **Sejil Nooh Ansari (1BM23IS221)** and **Shlok Naik Charan (1BM23IS229)** in partial fulfilment for the award of degree of Bachelor of Engineering in **Information Science and Engineering** from **Visvesvaraya Technological University, Belgaum** during the year **2025-2026**. It is certified that all corrections/suggestions indicated for Internal Assessments have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering Degree.

Dr. Shobha T
Associate Professor

Dr. M K Nalini
Professor and HOD

Dr. Bheemsha Arya
Principal

Examiners

Name of the Examiner

Signature of the Examiner

1.

2.



DECLARATION

Samanyu Gupta (1BM23IS216), Sarvagnya Akamanchi (1BM23IS220), Sejl Nooh Ansari (1BM23IS221) and Shlok Naik Charan (1BM23IS229) students of B.E. Information Science and Engineering, B.M.S. College of Engineering, Bangalore - 19, hereby declare that the capstone project entitled **“DryRunAI: A GAN-Centric Generative AI Framework for Pre-Launch Product Simulation and Product–Market Fit Optimization”** is an authentic work carried out under the supervision and guidance of **Dr. Shobha T, Associate Professor**, Department of Information Science and Engineering, B.M.S. College of Engineering, Bangalore. We have not submitted the matter embodied to any other university or institution for the award of any other degree.

Place : Bengaluru

Date : 5th May 2026

Name	USN	Signature
Samanyu Gupta	1BM23IS216	
Sarvagnya Akamanchi	1BM23IS220	
Sejl Nooh Ansari	1BM23IS221	
Shlok Naik Charan	1BM23IS229	

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Samanyu Gupta (1BM23IS216)

Sarvagnya Akamanchi (1BM23IS220)

Sejil Nooh Ansari (1BM23IS221)

Shlok Naik Charan (1BM23IS229)

ABSTRACT

Product launches involve significant uncertainty due to unpredictable customer perception and rapidly changing market dynamics. Traditional evaluation approaches such as surveys and post launch analytics fail to proactively assess risk before deployment. While Generative AI tools have gained popularity, most applications focus on content generation rather than simulation driven decision support. This project proposes DryRun AI, a GAN-centric pre-launch simulation platform designed to evaluate product and marketing strategies before market release. The core of the system is a Generative Adversarial Network (GAN) framework that generates realistic synthetic product visuals and controlled variations across attributes such as color, texture, layout, and presentation style.

By exploring the latent space of GAN models, the system produces multiple optimized visual alternatives that simulate potential real-world representations of a product. These synthetic variants are evaluated under multiple launch scenarios using rule-based simulation and LLM-driven persona reasoning to approximate audience reactions. The platform integrates optimization techniques such as latent space manipulation, hyperparameter tuning, scenario weighting, and threshold-based risk scoring to identify configurations that maximize engagement likelihood while minimizing predicted risk. Unlike traditional marketing tools, DryRun AI positions GANs not merely as content generators but as a decision-support engine for strategic simulation. The system demonstrates how generative modeling combined with optimization-driven evaluation can reduce launch uncertainty, improve strategic planning, and provide measurable pre-launch insights.

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CHAPTER 1

INTRODUCTION

1.1 Overview

In today’s highly competitive and technology-driven market environment, launching a successful product has become increasingly complex. Organisations invest substantial amounts of time, capital and resources into product development, branding, packaging, advertising and market research before introducing products to consumers. Despite these efforts, a large percentage of products fail after launch due to poor customer reception, weak branding strategies, pricing issues and incorrect assumptions regarding consumer behaviour.

According to multiple industry reports, nearly 70–80 percent of newly launched products fail within their first year due to inadequate market understanding and weak product-market alignment. Start-ups frequently struggle because they launch products without sufficient validation, while larger organisations often lose millions due to failed product campaigns.

Traditional product validation methods such as surveys, interviews, focus groups and A/B testing are commonly used to understand customer behaviour. Although these techniques provide useful insights, they often suffer from limitations such as high operational costs, delayed feedback cycles and human bias. These methods also fail to support large-scale experimentation involving multiple product variations.

Modern consumers are influenced by rapidly changing trends, social media platforms, competitor activity and dynamic pricing strategies. Their preferences change quickly, making traditional validation systems less effective in predicting product success.

Recent advancements in Artificial Intelligence have created opportunities to solve these problems through intelligent simulation systems. Technologies such as Generative Adversarial Networks (GANs), Large Language Models (LLMs), Reinforcement Learning

and Agent-Based Systems are transforming how businesses analyse customer behaviour and make strategic decisions.

Generative Adversarial Networks can generate realistic product variations by learning patterns from existing datasets. These models can create synthetic product designs, packaging styles and promotional creatives that can be evaluated before launch.

Large Language Models can simulate human-like reasoning and behavioural responses. They can represent multiple customer personas and provide feedback regarding product appeal, emotional response, pricing perception and potential risks.

DryRunAI is a GAN-centric Generative AI framework designed to help organisations simulate product launches before entering real markets. The platform enables businesses to test product concepts, visuals and marketing strategies virtually before making expensive real-world investments.

The system generates multiple product variations and evaluates them using simulated customer personas. The feedback obtained from these personas is used to improve future product designs through iterative optimisation.

DryRunAI reduces uncertainty in product development and enables organisations to make more informed business decisions.

1.2 Motivation

Product failures continue to remain a significant challenge across multiple industries. Every year, thousands of products fail because organisations misjudge consumer demand, launch ineffective branding campaigns or fail to understand customer expectations.

Start-ups face particularly high risks because they often operate with limited budgets and cannot afford repeated product failures. Established organisations may also suffer heavy financial losses when large-scale product launches fail.

Traditional validation methods present several challenges.

- Surveys often produce biased responses because customers may not accurately communicate their true preferences.
- Focus groups involve small sample sizes that may not represent broader markets.
- A/B testing requires large user traffic and is generally conducted after product launch.
- Manual testing processes consume considerable time and financial resources.
- Traditional systems cannot easily test multiple product variations simultaneously.

Modern markets are highly dynamic. Consumer preferences are influenced by social media trends, influencer marketing, seasonal demand and competitor launches.

Businesses require faster and more scalable validation systems that can simulate customer reactions before launch.

Artificial Intelligence provides an effective solution by enabling automated simulation, predictive analysis and large-scale experimentation.

The motivation behind DryRunAI is to create a platform that reduces uncertainty, lowers operational costs and helps organisations make better product launch decisions.

1.3 Objective

The primary objective of DryRunAI is to develop an intelligent pre-launch simulation platform that estimates product-market fit using Generative AI technologies.

The specific objectives of the project are:

- To generate realistic synthetic product designs using Generative Adversarial Networks.
- To simulate customer behaviour using Large Language Model-based personas.

- To analyse customer sentiment and behavioural responses.
- To identify product risks before launch.
- To optimise product designs using simulated feedback loops.
- To reduce financial losses caused by failed launches.
- To improve product decision-making efficiency.
- To provide businesses with actionable recommendations before launch.

The project aims to combine technical innovation with practical business applications.

1.4 Scope

The scope of DryRunAI extends across multiple industries where product appearance, branding and customer perception significantly influence purchasing behaviour.

The system can be used in:

- E-commerce platforms
- Consumer goods companies
- Fashion brands
- Software as a Service organisations
- Marketing agencies
- Retail businesses
- Start-ups

The platform can be used for:

- Product design validation
- Packaging optimisation
- Marketing campaign evaluation
- Pricing experimentation
- Customer sentiment simulation
- Risk analysis

The current implementation focuses on proof-of-concept development using public datasets and predefined personas.

Future enhancements may include:

- Real-time customer data integration
- Multilingual persona simulations
- Regional consumer modelling
- Reinforcement learning optimisation
- Enterprise-level deployment

DryRunAI provides long-term scalability for both academic research and commercial implementation.

1.5 Existing System

Traditional product validation systems rely heavily on manual market research methods to understand customer preferences before launching products into the market. These systems are commonly used by organisations to analyse customer behaviour, validate product ideas and reduce risks associated with product launches. However, these methods often lack

scalability and fail to provide accurate real-time insights. The commonly used traditional validation methods include:

- **Surveys**

Surveys are one of the most widely used techniques for collecting customer feedback regarding product features, pricing preferences and purchasing behaviour. Companies distribute questionnaires to potential users and analyse their responses to identify customer needs.

Although surveys provide structured feedback, they often suffer from response bias because customers may not accurately communicate their true preferences. Many participants may also provide incomplete responses or may not represent the broader target audience.

- **Focus Groups**

Focus groups involve a small number of participants discussing product ideas under the supervision of a moderator. These discussions help companies understand emotional reactions, preferences and product expectations.

However, focus groups often suffer from groupthink behaviour where dominant participants influence the opinions of others. Moderator bias may also affect the quality of feedback. Since only a limited number of participants are involved, the results may not represent larger markets.

- **A/B Testing**

A/B testing is commonly used to compare two versions of a product, advertisement or website design to determine which version performs better.

This method requires significant user traffic to generate meaningful results and is generally conducted after product launch. As a result, businesses often identify problems only after investing heavily in product deployment.

- **Beta Testing**

Beta testing involves releasing products to a limited group of users before full-scale launch. These users provide feedback regarding usability, performance and product features.

Although beta testing helps identify product issues, it often generates delayed feedback and fails to simulate large-scale market behaviour.

- **Social Media Analysis**

Many organisations analyse customer reviews, online discussions and social media comments to understand public sentiment regarding products.

This method often produces unstructured feedback that may be difficult to analyse systematically. Social media trends may also fluctuate rapidly, making long-term decision-making difficult.

Limitations of Existing Systems include:

- High operational costs
- Slow feedback cycles
- Human bias in decision-making
- Limited scalability
- Delayed optimisation opportunities
- Inability to test multiple product variations simultaneously
- Poor prediction of customer behaviour
- Higher product launch risks

Due to these limitations, traditional product validation systems are often inefficient in today's rapidly changing business environment. Organisations require faster, smarter and more scalable alternatives that can simulate customer reactions before launching products.

DryRunAI addresses these limitations by introducing Artificial Intelligence-based simulation techniques that provide faster feedback, improved scalability and better product-market fit estimation.

1.6 Proposed System

DryRunAI introduces an Artificial Intelligence-driven alternative to traditional product validation systems by enabling organisations to simulate product launches before entering the real market. Unlike conventional approaches that depend on physical surveys, focus groups and post-launch A/B testing, the proposed system uses Generative AI to create product variations and evaluate them through simulated customer personas.

The proposed system is designed as a closed-loop optimisation framework that continuously improves product designs based on simulated market feedback. It combines the strengths of Generative Adversarial Networks for visual generation and Large Language Models for behavioural simulation.

The working of the proposed system is divided into the following stages:

- **Stage 1: Product Input**

The organisation uploads product images, product descriptions, branding details and marketing campaign information into the system. Users may also define target customer segments and launch scenarios.

- **Stage 2: GAN-Based Product Generation**

The GAN model generates multiple synthetic product design variations by modifying product attributes such as colour, packaging style, layout, aesthetics and branding elements.

- **Stage 3: Scenario Configuration**

The system allows users to create multiple launch scenarios based on factors such as pricing strategy, branding style, target demographics and promotional messaging.

- **Stage 4: LLM-Based Audience Simulation**

Large Language Model-based personas simulate customer reactions by analysing generated product visuals. These personas represent different customer demographics, behavioural patterns and purchasing preferences.

- **Stage 5: Sentiment and Risk Analysis**

The system evaluates customer sentiment, identifies risks, detects possible misinterpretations and analyses engagement likelihood.

- **Stage 6: Feedback-Driven Optimisation**

The feedback generated from simulated audiences is used as an optimisation signal to improve future product variations generated by the GAN model.

- **Stage 7: Final Recommendation**

The system compares multiple product variations and recommends the most suitable product design for market launch.

The proposed system offers several advantages over traditional validation systems:

- Reduces product launch uncertainty
- Minimises financial losses
- Improves product-market fit estimation
- Enables faster decision-making
- Supports scalable testing across multiple scenarios
- Reduces dependency on manual research methods

As evident in the below figure 1.6 DryRunAI provides organisations with a virtual testing environment where product ideas can be validated efficiently before real-world implementation.



Fig 1.6 Proposed System Blueprint that lays out the features and user interface

CHAPTER-2

PROBLEM STATEMENT

2.1 Problem Statement

In modern business environments, product launches involve significant uncertainty due to unpredictable customer behaviour, rapidly changing market trends and increasing competition. Organisations invest large amounts of money in product development, branding, manufacturing, packaging and marketing campaigns before introducing products to consumers. Despite these investments, many products fail after launch because organisations are unable to accurately predict customer reactions before entering the market.

A major challenge faced by businesses is the lack of effective pre-launch validation systems that can simulate real-world customer behaviour. Traditional market research techniques such as surveys, focus groups, beta testing and A/B testing often provide delayed, expensive and limited feedback. These traditional systems suffer from several major problems:

- Limited audience reach
- High operational costs
- Slow feedback collection
- Human bias in responses
- Inability to test multiple product variations simultaneously
- Delayed design optimisation
- Poor scalability
- Lack of real-time decision-making support

In many cases, businesses identify product flaws only after launching products in real markets. At this stage, correcting design issues, branding failures or pricing mistakes becomes expensive and time-consuming.

Start-ups face even greater challenges because they often operate with limited financial resources and cannot afford repeated failures. Large organisations may also suffer reputational damage when unsuccessful product launches negatively impact customer trust.

Existing Generative AI systems primarily focus on content generation rather than decision simulation. Most current AI tools generate text, images or advertisements but do not simulate customer reactions or optimise products based on behavioural feedback.

There is currently no effective system that combines Generative Adversarial Networks, Large Language Models and feedback-driven optimisation to simulate product launches before real-world deployment.

DryRunAI addresses this problem by providing an intelligent simulation platform that enables organisations to test product concepts, analyse customer reactions and optimise products before launch.

2.2 Motivation

The motivation behind this project arises from the growing number of unsuccessful product launches across industries. Businesses frequently fail because they rely on assumptions rather than data-driven decision-making.

Modern markets are influenced by:

- Social media trends

- Rapid competitor launches
- Changing consumer expectations
- Price sensitivity
- Brand perception
- Seasonal demand fluctuations

Traditional research techniques are unable to adapt quickly to these changes.

Businesses require systems that can:

- Generate product variations rapidly
- Simulate diverse customer behaviour
- Analyse market risks
- Reduce development costs
- Improve decision-making speed

Generative AI technologies provide an opportunity to solve these problems by automating simulation and optimisation processes.

The motivation behind DryRunAI is to create a scalable platform that enables organisations to validate products intelligently before entering competitive markets.

2.3 Objectives

The major objectives of the project are:

- **To generate synthetic product variations** using Generative Adversarial Networks for design experimentation.

- **To simulate customer personas** using Large Language Models to analyse behavioural responses.
- **To identify product risks** before market launch.
- **To evaluate multiple launch scenarios** involving pricing, branding and marketing strategies.
- **To improve product-market fit estimation** through simulation-based feedback.
- **To reduce financial losses** caused by unsuccessful product launches.
- **To create a scalable framework** that can be extended across multiple industries.

The overall objective of DryRunAI is to create an intelligent decision-support platform that enables safer, faster and more efficient product launches.

CHAPTER-3

DETAILED SURVEY

3.1 Overview of Literature Survey

The development of DryRunAI requires the integration of multiple Artificial Intelligence domains including Generative Adversarial Networks, Large Language Models, persona simulation frameworks, autonomous AI agents, reinforcement learning systems and product design optimisation frameworks. Since the proposed system combines multiple technologies, an extensive literature survey was conducted to understand existing research contributions and identify research gaps.

A total of **44 research papers** were analysed across various domains relevant to the project. These papers were selected based on their relevance to product generation, behavioural simulation, optimisation systems and AI-driven decision-making.

The literature survey helps in understanding:

- Existing methodologies used for synthetic product generation
- Techniques used for customer persona simulation
- Behaviour modelling approaches
- Feedback optimisation systems
- Real-world product design applications
- Limitations of existing technologies
- Research gaps addressed by DryRunAI

The surveyed papers were classified into the following categories:

- GAN-based visual generation papers
- LLM persona simulation papers
- AI agent architecture papers
- Feedback optimisation papers
- Application-oriented

3.2 GAN-Based Visual Generation Papers

Generative Adversarial Networks (GANs) form the core of visual generation in DryRunAI by enabling realistic and diverse product design outputs. These models focus on generating high-quality images, transforming styles, and improving design creativity. Different GAN variants contribute unique capabilities such as conditional generation, image translation, and scalable synthesis. However, most existing works lack integration with customer behavior simulation and iterative feedback mechanisms.

- **DCG-GAN: Design Concept Generation with GANs (Ghasemi et al., 2024)**
This paper proposes a GAN-based framework focused on generating innovative product design concepts while ensuring diversity and desirability. It introduces evaluation metrics to assess the quality and novelty of generated outputs. The model is useful for creative design exploration but does not incorporate customer feedback or optimization loops. Its primary limitation lies in its isolation from real-world consumer simulation.
- **StyleGAN: A Style-Based Generator Architecture (Karras et al., 2019)**
StyleGAN introduces a style-based architecture that enables high-resolution image generation with fine-grained control over visual attributes. It allows manipulation of latent features to generate diverse and realistic product variations. This makes it highly suitable for product visualization tasks in DryRunAI. However, the model requires significant computational resources, limiting its scalability in low-resource environments.
- **Image-to-Image Translation with Conditional GAN (Phillip Isola et al., 2017)**
This research focuses on generating product prototypes while also predicting emerging design trends using GANs. It combines visual generation with trend analysis, making it useful for market-driven product development. The model helps bridge the gap between design and consumer preference forecasting. However, it still lacks direct integration with user behavior.
- **CycleGAN (Jun-Yan Zhu, Taesung Park, and Phillip Isola, et al., 2017)**
CycleGAN enables unpaired image-to-image translation, allowing transformation of product styles without requiring aligned datasets. It is widely used for redesigning product packaging and visual aesthetics. This flexibility makes it valuable for simulating different branding strategies. However, it does not generate entirely new designs from scratch.

- **Pix2Pix GAN (Phillip Isola, Jun-Yan Zhu, Alexei A. Efros, et al., 2017)**
Pix2Pix is a conditional GAN that performs image-to-image translation using paired datasets. It is effective in transforming sketches into realistic product images or modifying existing designs. This makes it suitable for controlled product redesign simulations. Its dependency on labeled data limits its applicability in large-scale scenarios.
- **CreativeGAN (Ahmed Elgammal, Bingchen Liu, Mohamed Elhoseiny, et al., 2021)**
CreativeGAN focuses on enhancing the creativity of generated outputs by introducing novel patterns and design variations. It improves the diversity of generated content beyond standard GAN outputs. This is useful for generating unique product concepts in DryRunAI. However, evaluation of creativity remains subjective and difficult to quantify.
- **Deep Convolutional GAN (Alec Radford, Luke Metz, Soumith Chintala, et al. 2016)**
DCGAN introduced stable training mechanisms using convolutional architectures for GANs. It significantly improved the reliability of image generation and became a foundational model for later GAN variants. Its simplicity and effectiveness make it useful for baseline implementations. However, it lacks the advanced control features seen in newer models.
- **StyleGAN2 Improvements (Tero Karras, Samuli Laine, and Timo Aila et al., 2020)**
StyleGAN2 improves upon the original StyleGAN by reducing artifacts and enhancing image realism. It provides more stable training and better-quality outputs. This makes it suitable for high-fidelity product visualization. Despite improvements, computational cost remains a challenge.
- **Conditional GAN (Mehdi Mirza and Simon Osindero, 2014)**
Conditional GAN introduces the concept of guiding generation using input conditions such as labels or attributes. This enables controlled product generation based on specific requirements. It is particularly useful for customizing designs according to target audiences. However, its performance depends heavily on the quality of conditioning data.
- **InfoGAN (Xi Chen, Yan Duan, and Rein Houthoofd et al., 2016)**
InfoGAN enhances interpretability by learning disentangled latent representations. It allows better control over specific visual features such as shape, color, or texture. This

improves explainability in generated outputs. However, it requires careful tuning to achieve meaningful feature separation.

- **BigGAN (Andrew Brock, Jeff Donahue, and Karen Simonyan et al., 2019)**
BigGAN focuses on large-scale image generation with high quality and diversity. It demonstrates the ability to generate detailed and realistic images across multiple categories. This makes it suitable for scalable product generation systems. However, it demands extremely high computational resources.

3.3 Large Language Model Persona Simulation Papers

Large Language Models (LLMs) enable DryRunAI to simulate customer behavior, opinions, and decision-making processes. These models generate realistic textual feedback based on predefined personas and scenarios. They help in understanding how different customer segments might react to a product. However, most systems lack integration with visual generation and iterative feedback loops.

- **Social Simulacra (Nikhil Sharma, Ingrid Haque, David Crandall et al., 2023)**
This paper demonstrates how LLMs can simulate customer responses using structured persona prompts. It allows generation of realistic feedback based on demographic and behavioral traits. This approach is useful for early-stage product evaluation. However, it may lack consistency across multiple simulation runs.
- **Quantifying Persona Effects in LLM Simulations (Hu et al., 2024)**
This research introduces methods to measure the impact of persona variables on generated responses. It improves realism by making outputs more aligned with defined user profiles. The study provides insights into how personas influence decision-making. However, it still lacks integration with real-world feedback data.
- **Large Language Models as Simulated Economic Agents (Horton, 2023)**
This paper explores the use of LLMs as economic agents capable of making purchasing decisions. It models consumer behavior in simulated marketplaces. This is useful for predicting product success in different economic scenarios. However, it simplifies complex human decision-making processes.

- **Persona Prompting for LLM Behaviour (Zhou et al., 2024)**
This work focuses on designing prompts that represent different demographic and psychological profiles. It enables targeted simulation of specific customer segments. This improves the relevance of generated feedback. However, results depend heavily on prompt engineering quality.
- **Scaling Laws in LLM Personality Simulation (Bai et al., 2025)**
This paper studies how increasing model size improves personality consistency and realism. It shows that larger models generate more coherent and believable persona-based responses. This enhances simulation accuracy in DryRunAI. However, larger models require more computational resources.
- **Dynamic Persona Refinement Framework (Yao et al., 2025)**
This framework introduces iterative refinement of personas based on feedback loops. It allows continuous improvement of simulation accuracy over time. This is useful for adaptive product testing environments. However, it increases system complexity.

3.4 AI Agent Architecture Papers

AI agent architectures enable automation and decision-making by combining memory, reasoning, and interaction capabilities. These systems simulate human-like behavior over time and can manage complex workflows. They play a key role in integrating different components of DryRunAI. However, most existing systems focus on isolated functionalities.

- **PersonaAgent Framework (conceptual framework inspired by Zhang et al., 2018)**
This framework creates personalized AI agents tailored to specific user profiles. It enables simulation of individual behaviors and preferences. This is useful for personalized product feedback generation. However, scalability remains a challenge.
- **User Behaviour Simulation with LLM Agents (Wang et al., 2025)**
This research uses LLM-based agents to simulate realistic user interactions. It models how users respond over multiple interactions. This improves behavioral realism in simulations. However, it lacks integration with visual systems.
- **Stanford Generative Agents (Joon Sung Park, Joseph C O'Brien et al., 2023)**
This paper introduces agents with memory, planning, and reflection capabilities. These

agents can simulate long-term human behavior in virtual environments. This is useful for continuous product interaction simulation. However, it requires significant computational resources.

3.5 Feedback Optimisation Papers

Feedback optimization ensures that generated outputs improve over time based on user responses and evaluations. These methods use reinforcement learning and human feedback to refine model outputs. They are essential for achieving product-market fit. However, they are often implemented separately from generation systems.

- **Deep Reinforcement Learning for Dialogue Generation (Jiwei Li et al., 2016)**
This approach uses reinforcement learning to improve persona-based outputs over multiple iterations. It refines responses based on previous feedback. This enhances the quality and consistency of simulations. However, it requires careful reward design.
- **RL with Human Feedback for LLM Optimisation (Paul Christiano et al., 2017)**
Reinforcement Learning with Human Feedback improves model outputs using human evaluations. It aligns generated responses with user expectations. This is widely used in modern LLM systems. However, collecting high-quality feedback is resource-intensive.

3.6 Application-Based Papers

Application-based papers demonstrate how GANs and LLMs are used in real-world scenarios such as fashion design and product development. These studies validate the practical feasibility of combining AI techniques for creative tasks. They provide insights into implementation challenges and performance. However, most applications focus on specific domains rather than generalized systems like DryRunAI.

3.7 Comparative Analysis of Existing Systems

Existing systems in literature are often specialized, focusing either on image generation, text simulation, or automation. GAN-based systems generate visuals, while LLM-based systems simulate user responses independently. Agent systems handle workflows, and feedback systems

optimize outputs separately. There is a clear lack of integration across these components, limiting their effectiveness in complete product simulation.

3.8 Research Gap Analysis

Current research lacks a unified system that integrates visual generation, persona simulation, agent-based interaction, and feedback optimization. There is limited work on pre-launch product simulation and product-market fit analysis using AI. Most systems do not support iterative redesign based on simulated feedback. DryRunAI addresses these gaps by combining all components into a single, scalable framework.

3.9 Conclusion of Literature Survey

The literature survey conducted for DryRunAI shows that significant research has been carried out in individual domains such as Generative Adversarial Networks, Large Language Models, AI agent systems and feedback optimisation frameworks. These technologies have demonstrated strong capabilities in solving specific problems related to image generation, behavioural simulation and automation.

GAN-based research papers such as StyleGAN, StyleGAN2, DCG-GAN, CycleGAN and Pix2Pix demonstrate strong capabilities in generating realistic product visuals and design variations. These systems provide the visual generation foundation required for DryRunAI.

Similarly, LLM-based research papers demonstrate that Artificial Intelligence systems can successfully simulate customer behaviour, purchasing decisions and emotional reactions through persona modelling techniques.

AI agent frameworks such as AutoGPT, LangChain and Stanford Generative Agents contribute towards workflow automation and multi-step reasoning capabilities.

Feedback optimisation research highlights how iterative feedback loops can continuously improve Artificial Intelligence outputs over time.

Although these technologies are highly effective individually, most existing systems operate

in isolation and solve only one specific part of the problem.

The major limitations identified in current systems include:

- Lack of complete product launch simulation systems
- Lack of integration between GANs and LLMs
- Limited feedback-driven optimisation
- Poor product-market fit estimation
- Lack of real-world commercial simulation platforms

DryRunAI addresses these gaps by combining visual generation, customer simulation and optimisation into one unified framework.

The literature survey validates both the novelty and technical feasibility of the proposed system and provides a strong foundation for implementation in the upcoming chapters.

CHAPTER-4

SURVEY SUMMARY TABLE

4.1 Overview

The literature survey presented in Chapter 3 analysed various research papers related to Generative Adversarial Networks, Large Language Models, AI agents and feedback optimisation systems. These papers were studied to understand existing methodologies, technological advancements and limitations in current systems.

Since DryRunAI integrates multiple Artificial Intelligence components, it is important to summarise the major findings of the literature survey in a structured format. The survey summary table helps compare the selected research papers based on their domain, technologies used, contributions and limitations.

The table provides a clear overview of how each research paper contributes to the development of DryRunAI.

4.2 Purpose of Survey Summary Table

The survey summary table was created to achieve the following objectives:

- To compare multiple research papers in a structured manner
- To identify the technologies used in existing systems
- To analyse the strengths of current research
- To identify limitations in existing solutions
- To determine how each paper contributes to DryRunAI
- To identify research gaps for future development

This structured comparison helps simplify the large amount of literature analysed in the previous chapter.

4.3 Literature Survey

The below table 4.3 shows detailed analysis of papers referred and provides a gist of approach the respective authors took along with the significant results they achieved that hold relevance to our proposed system.

Table 4.3: Literature Survey Summary Table

Sr. No.	Title of the Paper	Problem Addressed	Authors	Approach / Method	Results
1	DCG-GAN	Generating innovative product concepts	Ghasemi et al.	Uses a GAN architecture with dual-objective optimization—generator produces designs while discriminator evaluates both realism and novelty/desirability metrics using design evaluation functions.	Achieves high-quality, diverse product concepts with measurable creativity; balances innovation and practicality better than standard GANs.
2	GAN-Based Product Appearance Design	Predicting product design trends	Li et al.	Combines GAN + Deep CNN feature extraction trained on historical design datasets to learn evolving visual trends and generate future-oriented product designs.	Successfully predicts market-aligned design patterns, improving relevance of generated prototypes.
3	StyleGAN	High-quality image synthesis	Karras et al.	Introduces style-based generator where latent vectors control different visual features (texture, shape) through adaptive instance normalization (AdaIN).	Produces photorealistic, high-resolution images with fine-grained control over design attributes.
4	StyleGAN2	GAN instability and artifacts	Karras et al.	Improves architecture using weight demodulation, path length regularization, and removes artifacts caused by normalization.	Significantly improves image fidelity, stability, and training robustness.
5	CreativeGAN	Lack of creativity in generated outputs	Nobari et al.	Modifies GAN objective to reward deviation from training distribution using latent	Generates novel and unconventional designs beyond

				space exploration and creativity loss functions.	training data patterns.
6	CycleGAN	Need for unpaired image translation	Zhu et al.	Uses cycle-consistency loss to map images between two domains without paired datasets ($A \rightarrow B$ and $B \rightarrow A$).	Enables flexible design transformations without requiring labeled pairs.
7	Pix2Pix	Controlled image generation	Isola et al.	Conditional GAN trained with paired datasets, mapping input images to outputs using supervised learning.	Produces accurate and controllable transformations (e.g., sketches $\rightarrow$ product designs).
8	LLM Generated Persona Simulation	Simulating human-like feedback	Li et al.	Uses prompt-engineered personas (age, preferences, behavior traits) to guide LLM responses in simulating customer opinions.	Produces context-aware, human-like feedback, useful for virtual testing.
9	Quantifying Persona Effect in LLMs	Lack of measurable realism	Hu et al.	Introduces persona variables (demographics, preferences) and evaluates impact on LLM outputs using behavioral metrics.	Demonstrates significant improvement in realism and consistency of simulated users.
10	LLMs as Economic Agents	Modeling decision-making	Horton	Uses LLMs as agents in economic simulations, assigning roles and decision rules to mimic human economic behavior.	Shows LLMs can approximate human economic decisions in controlled experiments.
11	Stanford Generative Agents	Realistic virtual human simulation	Park et al.	Combines LLM + memory streams + planning + reflection modules to simulate long-term agent behavior.	Produces highly believable human-like agents capable of social interaction.
12	User Behavior Simulation with LLM Agents	Testing systems with users	Wang et al.	Builds multi-agent simulation environment where LLM agents interact dynamically using predefined behavioral goals.	Generates realistic interaction patterns, useful for system evaluation.
13	PersonaAgent Framework	Scalable persona modeling	Zhang et al.	Modular architecture integrating persona definition, memory, and reasoning modules for personalized agent creation.	Enables scalable and customizable persona agents across domains.

14	Population-Aligned Persona Generation	Lack of diversity in personas	Hu et al.	Uses population-level statistical modeling + LLM generation to create representative persona datasets.	Produces diverse and realistic population simulations.
15	Dynamic Persona Refinement	Static persona limitations	Yao et al.	Iteratively updates persona attributes using feedback loops and alignment mechanisms.	Improves accuracy and adaptability of personas over time.
16	PersonaGym Benchmark	Lack of evaluation standards	Samuel et al.	Introduces benchmark datasets, scoring metrics, and evaluation protocols for agents.	Provides standardized evaluation framework for comparing models.
17	Reliable LLM User Simulator	Inconsistent simulations	Li et al.	Enhances simulation using context retention, memory, and behavioral constraints.	Produces more stable and coherent simulated users.
18	ReAct	Separation of reasoning & acting	Yao et al.	Integrates chain-of-thought reasoning with action execution, allowing agents to think and act iteratively.	Improves decision accuracy and interpretability of agents.
19	RLHF	Poor alignment with human expectations	Ouyang et al.	Uses human feedback to train reward models, then applies reinforcement learning to optimize outputs.	Produces aligned, safe, and high-quality responses.
20	Multi-Turn Persona RL Optimization	Persona inconsistency over time	Abdulhai et al.	Applies reinforcement learning across multiple interactions to maintain persona consistency.	Achieves stable long-term persona behavior across conversations.

CHAPTER-5

SYSTEM REQUIREMENT SPECIFICATION

5.1 Introduction

System Requirement Specification (SRS) defines the functional and non-functional requirements of DryRunAI. It provides a detailed description of the system behaviour, technical requirements and operational expectations of the proposed platform.

The purpose of this chapter is to clearly define what the system should accomplish and the resources required for implementation. It helps developers understand the overall expectations of the platform before moving towards system design and development.

DryRunAI is designed as a Generative Artificial Intelligence-based simulation platform that helps organisations test product ideas before launching them into real markets. The system combines Generative Adversarial Networks, Large Language Models and optimisation frameworks to simulate customer reactions and improve product-market fit.

5.2 Functional Requirements

Functional requirements define the major operations that the system must perform.

The major functional requirements of DryRunAI include:

- **User Registration and Login** – The system should allow organisations to create accounts and securely log into the platform.
- **Product Upload Module** – Users should be able to upload product images, product descriptions and branding details.
- **Scenario Configuration Module** – Users should be able to configure pricing strategies, target demographics and launch scenarios.

- **GAN-Based Product Generation** – The system should generate multiple synthetic product variations.
- **Persona Simulation Module** – Large Language Models should simulate customer reactions based on predefined personas.
- **Sentiment Analysis Module** – The system should analyse customer responses as positive, negative or neutral.
- **Risk Detection Module** – The platform should identify possible product launch risks.
- **Feedback Optimisation Module** – The system should improve product designs using feedback loops.
- **Dashboard Module** – Users should be able to view reports, charts and simulation insights.
- **Final Recommendation Module** – The system should recommend the best product variation for launch.

The below table 5.2 briefly encapsulates the important functionalities of the system besides the expected outcome.

Table 5.2 Functional Requirements

Sr. No.	Module Name	Description	Inputs	Processing / Functionality	Outputs
1	User Registration and Login	Enables organisations to create accounts and securely access the system	User credentials (email, password, org details)	Authentication, validation, session management	Secure login access, user session
2	Product Upload Module	Allows users to upload product-related data	Product images, descriptions, branding details	Data validation, storage, preprocessing	Stored product data ready for simulation
3	Scenario Configuration Module	Enables users to define launch parameters	Pricing strategy, target demographic	Configuration setup, parameter mapping	Configured simulation environment

			s, launch scenario		
4	GAN-Based Product Generation	Generates multiple product variations	Uploaded product data, design inputs	GAN model generates synthetic variations using learned patterns	Multiple product design variants
5	Persona Simulation Module	Simulates customer behavior	Product variants, predefined personas	LLM-based persona reasoning and response generation	Simulated customer feedback
6	Sentiment Analysis Module	Analyzes customer responses	Persona feedback text	NLP-based sentiment classification (Positive/Negative/Neutral)	Sentiment scores and labels
7	Risk Detection Module	Identifies potential launch risks	Sentiment data, simulation results	Pattern detection, anomaly detection, rule-based + AI insights	Risk alerts and insights
8	Feedback Optimisation Module	Improves product design iteratively	Feedback data, risk insights	Reinforcement learning / feedback loop optimization	Refined product variations
9	Dashboard Module	Provides visual insights and reports	Simulation outputs, analytics data	Data aggregation, visualization (charts, graphs)	Interactive dashboard and reports
10	Final Recommendation Module	Suggests best product for launch	Optimized variants, analytics results	Ranking algorithms, scoring models	Recommended product variant

5.3 Non-functional Requirements

Non-functional requirements define the quality standards of the system. The major non-functional requirements include:

- **Performance** – The system should process simulations efficiently.
- **Scalability** – The platform should support multiple users simultaneously.
- **Security** – User information and uploaded product data should remain secure.
- **Reliability** – The system should provide consistent outputs.

- **Usability** – The platform interface should be user-friendly.
- **Maintainability** – Future updates should be easy to implement.
- **Availability** – The platform should remain accessible whenever required.
- **Portability** – The system should function across multiple devices and operating systems.

The following table 5.3 captures the key non functional requirements and their criteria.

Table 5.3 Non-Functional Requirements

Sr No	Requirement	Description	Metrics / Criteria	Implementation Approach	Expected Outcome
1	Performance	System should process simulations efficiently	Response time, latency, throughput	Use optimized models, GPU acceleration, parallel processing	Fast simulation execution with minimal delay
2	Scalability	System should handle multiple users simultaneously	Number of concurrent users, load handling capacity	Cloud infrastructure (AWS/Azure), load balancing, microservices	Smooth performance under high user load
3	Security	Protect user data and product information	Data encryption, authentication, authorization	SSL/TLS encryption, secure login (JWT/OAuth), database security	Safe and confidential data handling
4	Reliability	System should provide consistent outputs	System uptime, failure rate, error handling	Fault-tolerant design, redundancy, logging mechanisms	Stable and dependable system performance
5	Usability	Interface should be user-friendly	User satisfaction, ease of navigation, learning curve	Intuitive UI/UX design, dashboards, guided workflows	Easy-to-use platform with minimal training
6	Maintainability	System should support easy updates and fixes	Code modularity, documentation quality	Modular architecture, clean code practices, version control	Faster updates and easier debugging
7	Availability	System should be accessible at all times	Uptime percentage (e.g., 99.9%)	Cloud hosting, auto-scaling, backup systems	Continuous access without downtime
8	Portability	System should run on multiple platforms	Cross-platform compatibility	Web-based architecture, containerization (Docker)	Accessible across devices (PC, mobile, OS)

5.4 Hardware Requirements

The hardware requirements needed for implementing DryRunAI include:

- Processor: Intel i5/i7 or equivalent
- RAM: Minimum 8 GB
- Storage: Minimum 256 GB SSD
- GPU: NVIDIA GPU recommended for GAN processing

The above hardware configuration is sufficient for both development and simulation processes.

5.5 Software Requirements

The software requirements include:

- **Operating System:** Windows / Linux / macOS
- **Frontend Framework:** React / Streamlit
- **Backend Framework:** FastAPI / Flask
- **Programming Language:** Python
- **Database:** MySQL / MongoDB
- **GAN Frameworks:** PyTorch / TensorFlow
- Large Language Model APIs / Open-source LLMs
- Cloud deployment tools
- Visualisation libraries

These software tools as listed in table 5.5 support the implementation and deployment.

Table 5.5 Software Requirements

SL.No	Software Component	Options / Tools	Description	Purpose / Usage	Remarks
1	Operating System	Windows / Linux / macOS	Platform on which the system runs	Provides environment for development and deployment	Linux preferred for deployment
2	Frontend Framework	React / Streamlit	User interface development tools	Builds interactive dashboards and user interfaces	Streamlit faster for prototyping
3	Backend Framework	FastAPI / Flask	Server-side frameworks	Handles API requests, business logic, and model integration	FastAPI preferred for performance
4	Programming Language	Python	Core development language	Used for AI models, backend logic, and data processing	Widely used in AI/ML ecosystem
5	Database	MySQL / MongoDB	Data storage systems	Stores user data, product data, and simulation results	MongoDB useful for flexible data
6	GAN Frameworks	PyTorch / TensorFlow	Deep learning libraries	Used to build and train GAN models	PyTorch preferred for research flexibility
7	LLM Integration	APIs / Open-source LLMs	Language model access tools	Enables persona simulation and response generation	Can use OpenAI / HuggingFace models
8	Cloud Deployment Tools	AWS / Azure / GCP	Cloud platforms	Deploys and scales the application	Supports scalability and availability
9	Visualization Libraries	Matplotlib / Plotly / Seaborn	Data visualization tools	Generates charts, graphs, dashboards	Plotly useful for interactive visuals

5.6 Feasibility Analysis

The proposed system was analysed for feasibility before implementation.

- **Technical Feasibility** – The required technologies are available and practical for implementation.
- **Economic Feasibility** – Public datasets reduce operational costs.
- **Operational Feasibility** – Organisations can adopt the platform easily.

- **Legal Feasibility** – No private or proprietary datasets are used.
- **Schedule Feasibility** – The project can be completed within the academic timeline.

5.7 Conclusion

This chapter defined the major system requirements needed for implementing DryRunAI. These requirements provide a strong foundation for system development and implementation.

The next chapter discusses the detailed system design of the proposed platform.

CHAPTER-6

SYSTEM DESIGN

6.1 System Design

System design explains the overall structural framework of DryRunAI and describes how different modules interact with each other to perform product simulation, customer behaviour analysis and optimisation. The proposed system integrates multiple Artificial Intelligence technologies to create a unified pre-launch simulation platform.

The primary objective of system design is to ensure efficient communication between system components while maintaining scalability, flexibility and operational efficiency. The design focuses on reducing manual intervention and improving automation through Artificial Intelligence-based workflows.

DryRunAI follows a layered architecture where users interact with the front-end platform, while backend services coordinate Artificial Intelligence models, databases and optimisation engines. The major design objectives include:

- Improving scalability of product simulations
- Reducing processing delays
- Supporting multiple product variations
- Enabling customer persona simulation
- Supporting feedback-driven optimisation
- Providing final product recommendations

The system is designed to handle both current academic implementation requirements and future commercial scalability.

6.1.1 System Architecture

The system architecture of DryRunAI consists of multiple interconnected layers that work together to perform product simulation.

The major architectural components include:

- **User Interface Layer**

This layer allows users to register, log in, upload product details and configure simulation settings.

- **Application Layer**

This layer manages communication between the user interface and backend services.

- **GAN Processing Layer**

This layer generates synthetic product variations based on user inputs.

- **LLM Persona Layer**

This layer simulates customer reactions by generating behavioural feedback.

- **Optimisation Layer**

This layer refines product outputs based on feedback loops.

- **Database Layer**

This layer stores product data, user information and simulation results.

- **Analytics Dashboard Layer**

This layer presents final reports and recommendations.

These layers as shown clearly in Fig 6.1 work together to create a complete simulation environment that enables organisations to test products before launching them in real markets.

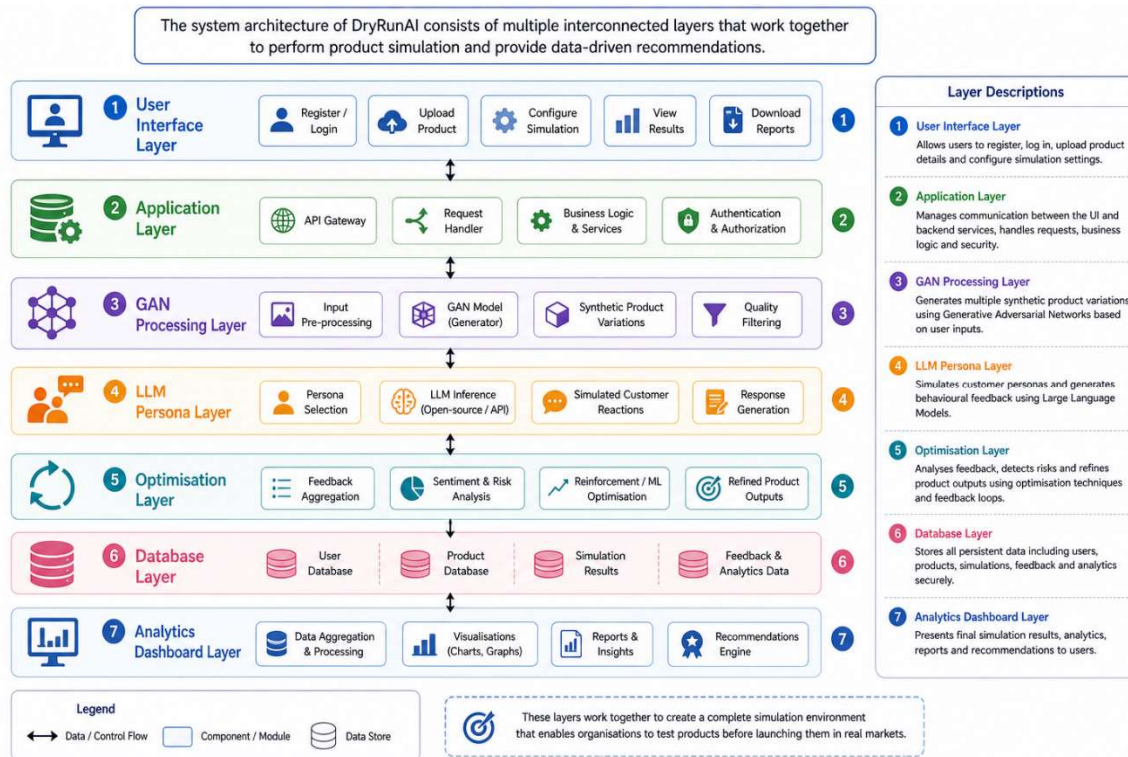


Fig 6.1 In-depth layer wise system architecture covering UI to backend operations.

6.1.2 Module Design

The DryRunAI platform consists of multiple modules that perform specific functions.

- **Authentication Module**

Handles user registration and login.

- **Product Upload Module**

Accepts product images, descriptions and branding information.

- **Scenario Configuration Module**

Allows users to define pricing strategies and target demographics.

- **GAN Generation Module**

Generates multiple synthetic product variations.

- **Persona Simulation Module**

Simulates customer reactions using Large Language Models.

• Sentiment Analysis Module

Analyses positive, negative and neutral responses.

• Risk Detection Module

Identifies potential product launch risks.

• Recommendation Module

Provides final recommendations.

• Dashboard Module

Displays simulation results.

Each module as seen here in Fig 6.2 performs specialised tasks that contribute to overall system efficiency.

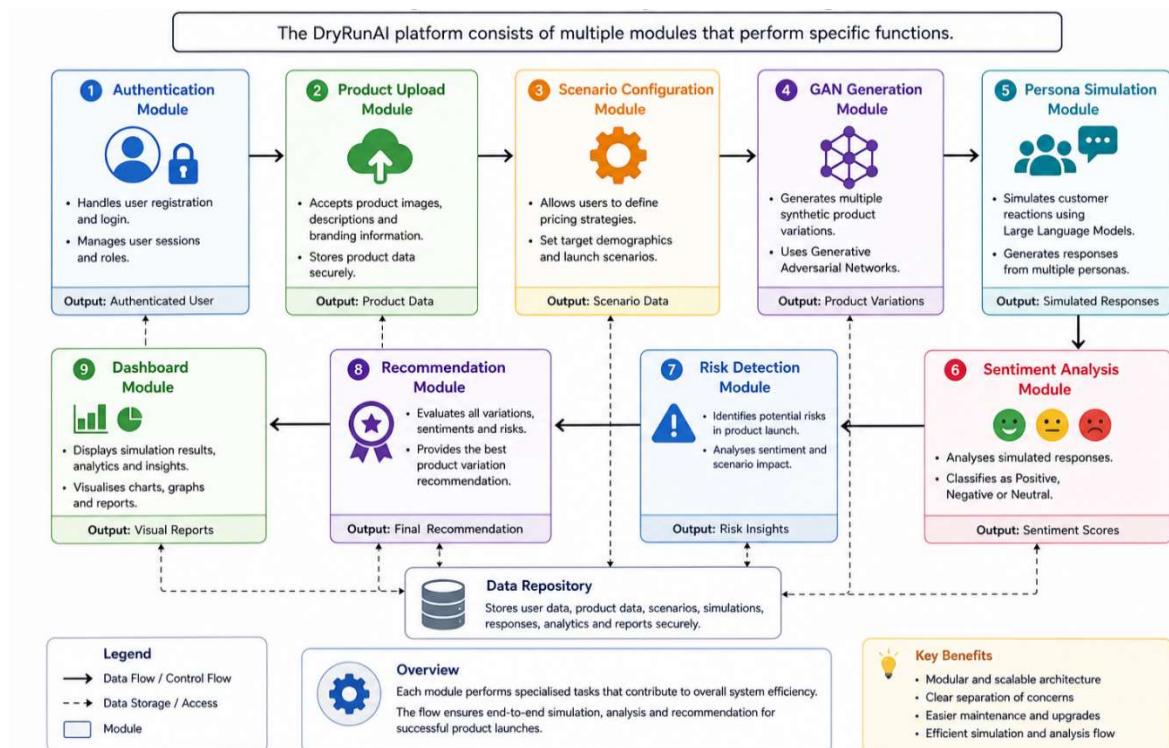


Fig 6.2 Module Design showing modular breakdown of the system with workflow.

6.2 Detailed Design

Detailed design explains the internal workflow and behavioural structure of the system. It focuses on interactions between users, modules and data entities.

This section explains the technical design diagrams used for implementation.

6.2.1 Class Diagram

The class diagram represents the object-oriented structure of DryRunAI.

The major classes include:

- User
- Product
- Simulation
- Persona
- Feedback
- Recommendation
- Admin

These classes interact with one another to support simulation workflows and output generation.

The class diagram as given in Fig 6.3 helps developers understand relationships between objects and supports easier implementation.

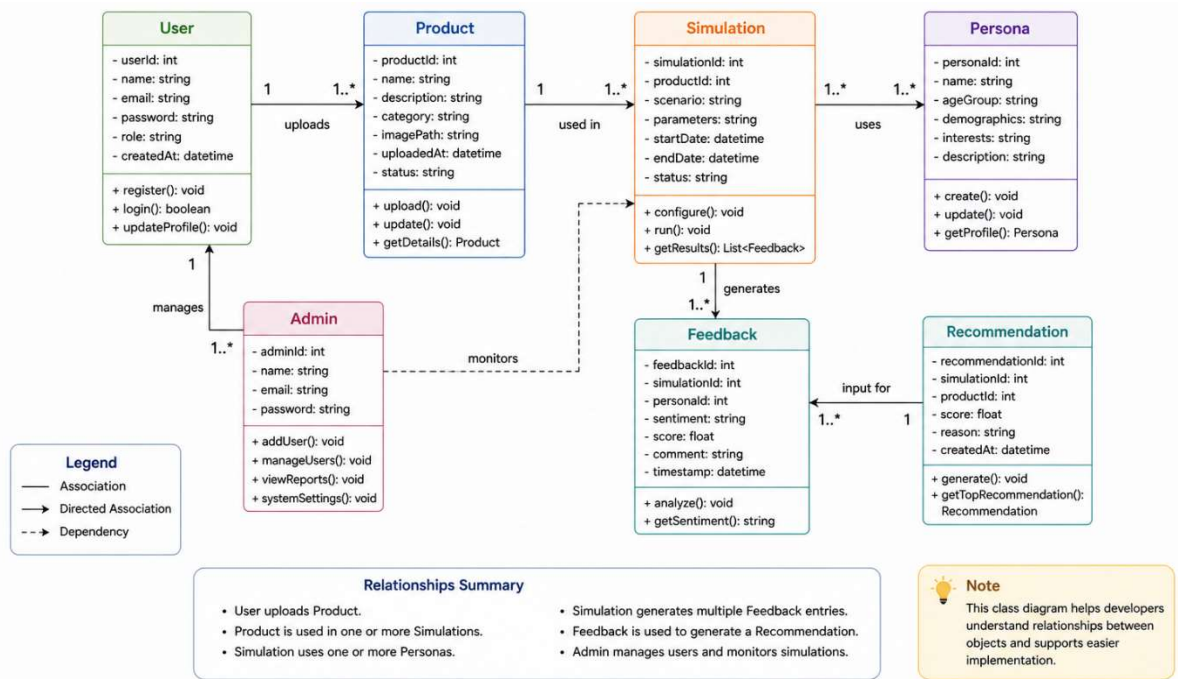


Fig 6.3 System’s class diagram enumerating required classes and their methods and attributes

6.2.2 Activity Diagram

The activity diagram explains the workflow followed during product simulation including :

- User login
- Product upload
- Scenario selection
- Product generation
- Persona simulation
- Feedback generation
- Risk analysis
- Final recommendation generation

This diagram Fig 6.4 explains operational flow and helps identify process dependencies.

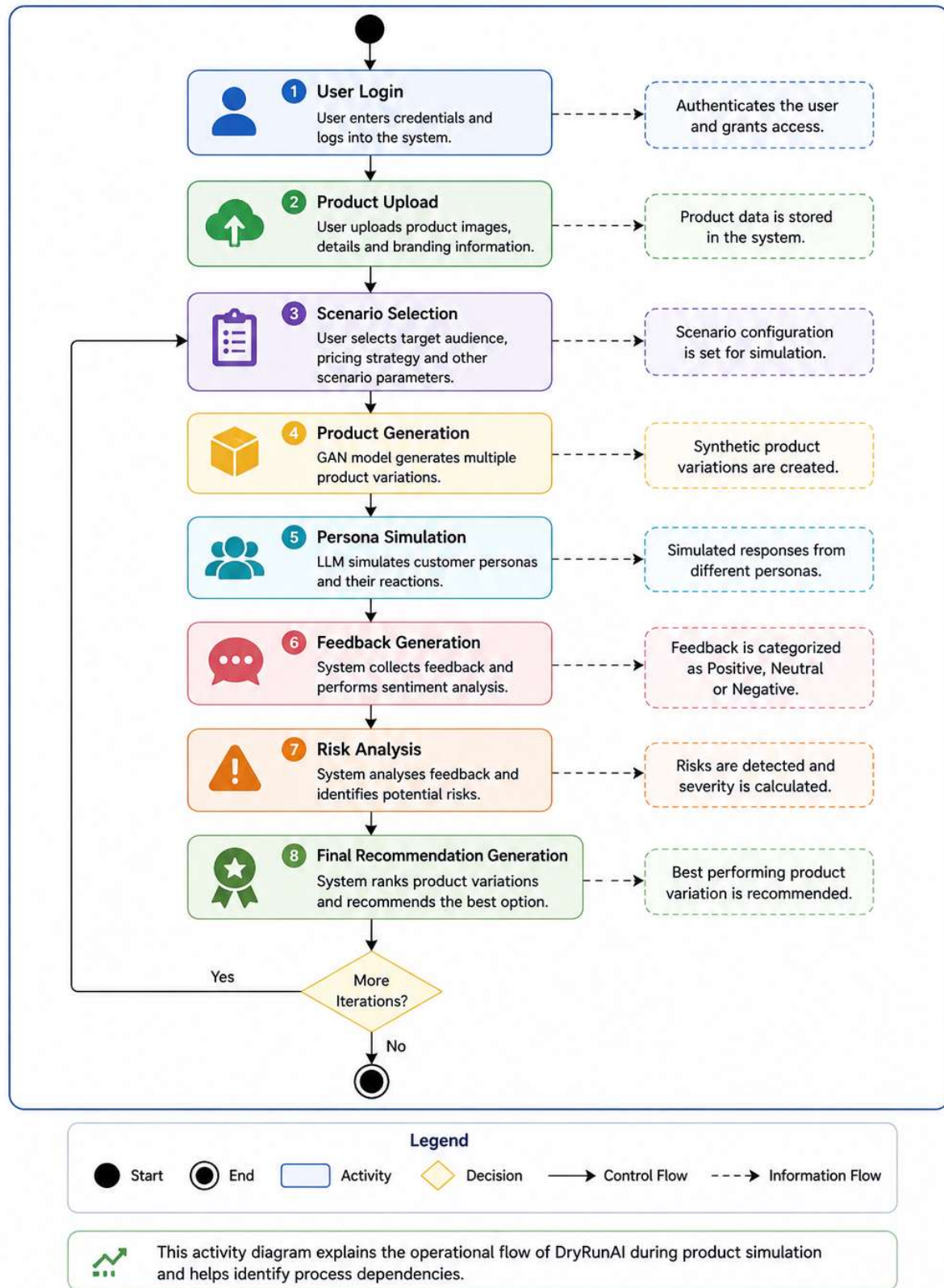


Fig 6.4 Activity Diagram of DryRun AI showcasing step by step internal processing

6.2.3 Use Case Diagram

The use case diagram below Fig 6.5 explains how users interact with the system. The major use cases include:

- Register
- Login
- Upload product
- Configure Scenarios
- Run Simulation
- View Analytics
- Download Report

This diagram helps identify all user-system interactions.

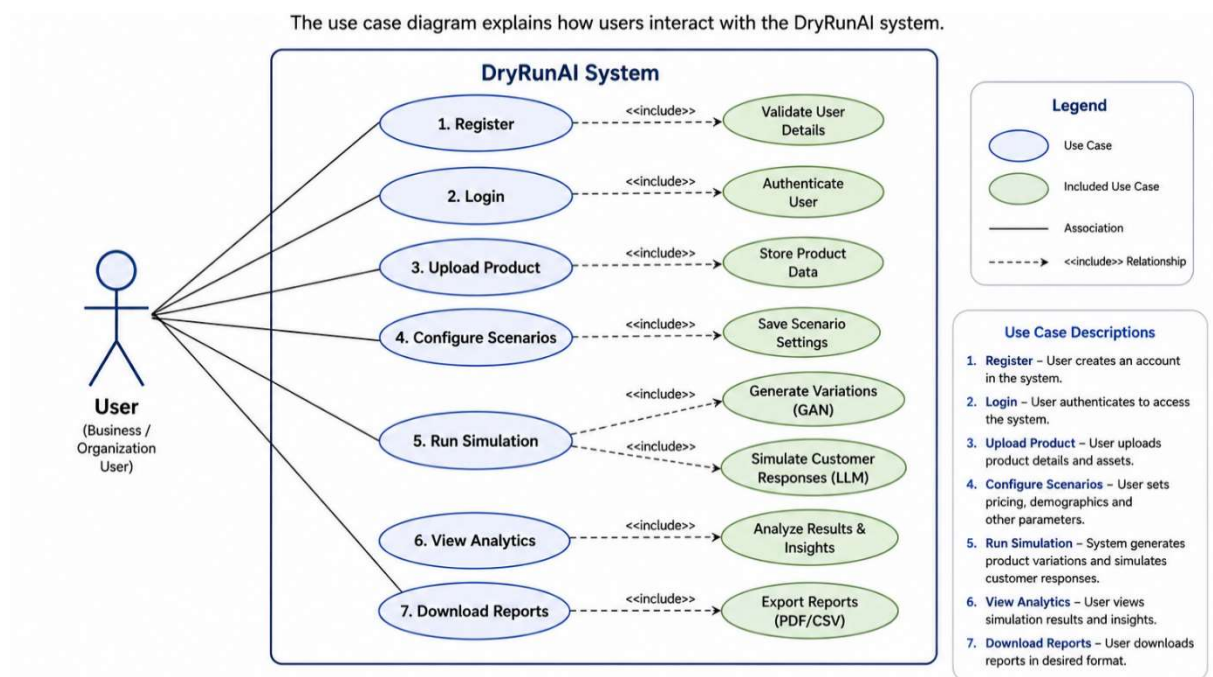


Fig 6.5 Use Case Diagram of DryRun AI showing abstract level interaction of end user

6.2.4 Scenarios

Scenarios as given in Fig 6.6 explain how DryRunAI can be used in real-world situations.

- **Scenario 1: Product Design Validation**

A company uploads product designs to test market reactions.

- **Scenario 2: Marketing Campaign Testing**

Users evaluate multiple advertisement creatives.

- **Scenario 3: Pricing Strategy Simulation**

Businesses test multiple pricing models.

- **Scenario 4: Risk Analysis**

The system identifies possible launch failures.

- **Scenario 5: Final Product Recommendation**

The platform recommends the best product variation.

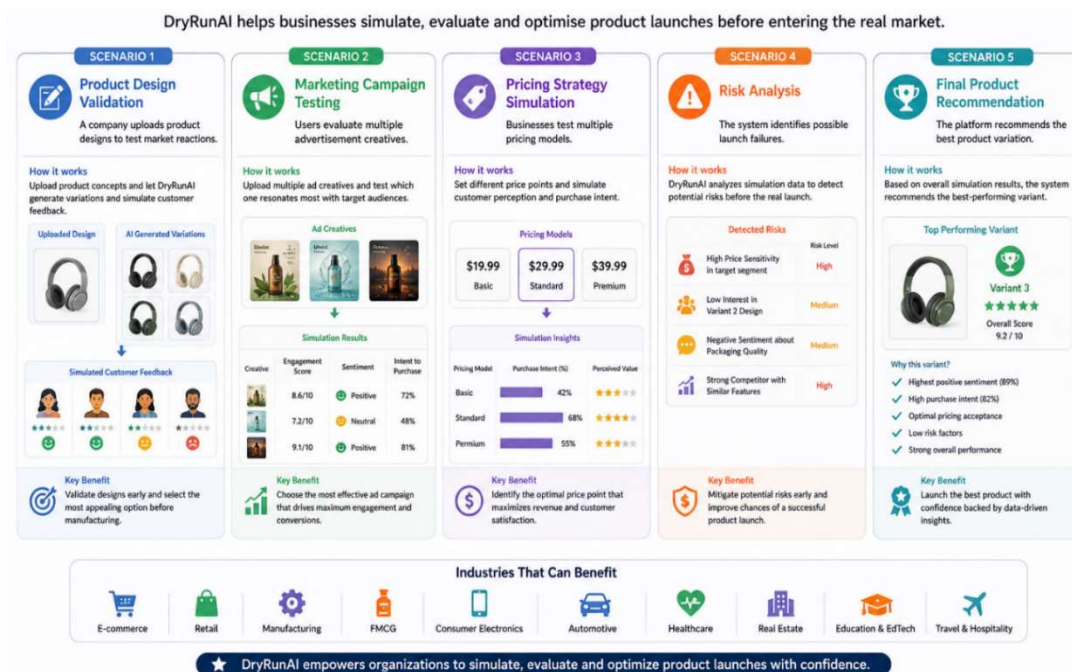


Fig 6.6 Practical implementation of DryRunAI across multiple industries.

CHAPTER-7

APPLICATION

7.1 Introduction

DryRunAI is designed to help organisations test product ideas before launching them into real markets. The platform can be applied across multiple industries where product design, branding and customer perception influence purchasing behaviour.

By simulating customer reactions and identifying risks before launch, the system helps organisations make better decisions and reduce financial losses.

7.2 Major Applications

The major applications of DryRunAI, as depicted in Fig 7.2, include:

- **E-Commerce** – Testing product visuals and packaging designs before launch.
- **Fashion Industry** – Analysing clothing designs and customer preferences.
- **Consumer Goods** – Evaluating branding and packaging strategies.
- **Marketing and Advertising** – Testing promotional campaigns and advertisements.
- **Software as a Service (SaaS)** – Validating digital products and pricing strategies.
- **Start-ups** – Reducing risks associated with new product launches.
- **Retail Businesses** – Understanding customer behaviour before product releases.



Fig 7.2 DryRun AI simulation across multiple industries

7.3 Future Scope

DryRunAI can be expanded into:

- Software-as-a-Service platforms
- Enterprise solutions
- Multi-domain simulations
- International market testing
- Real-time customer feedback integration

CONCLUSION

DryRunAI proposes a novel Generative Artificial Intelligence framework that helps organisations simulate product launches before entering real markets. The system combines Generative Adversarial Networks, Large Language Models and feedback-driven optimisation techniques to generate product variations, simulate customer behaviour and improve product-market fit.

The project addresses the limitations of traditional product validation methods by providing more scalable and cost-effective simulation capabilities. DryRunAI can be applied across industries such as e-commerce, fashion, marketing, consumer goods and Software as a Service platforms.

The proposed framework does not guarantee product success but helps organizations reduce uncertainty, identify risks and make better product decisions before launch. Future improvements may include real-time market integration, advanced reinforcement learning optimisation, multilingual persona simulation and large-scale commercial deployment.

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